Saudi Legal FC at AISA-ArabicFC: Disciplined Fine-Tuning and Gold-Free Post-Processing for Arabic Function Calling

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Abstract

We describe the Saudi Legal FC submission to the AISA-ArabicFC shared task on Arabic function calling (Nacar et al., 2026a). Our system is a parameter-efficient fine-tune of Qwen2.5-7B-Instruct (Team, 2025) with LoRA (Hu et al., 2022), trained with completion-only loss masking. Beyond the model, our central contribution is methodological: we use written, falsifiable score predictions and a validated local scorer to gate submissions, with a post-processing safety net that uses no gold at inference. On the public development set our best configuration reaches Argument Exact-Match (ArgEM) 0.828 and function-name accuracy 1.0 (Overall-B 0.914) under the official v1.3 evaluator. Its weights were then lost; the blind submission comes from a reconstruction selected against that model's frozen predictions, and ranks 12th on Track B with Overall-B 0.8946. We also report two negative findings—one retraining run, one rejected hypothesis—and an artifact-identity failure our behavioral checks caught before submission.

1 Introduction

Function calling—the interface through which a language model selects and parameterizes an external tool—has become central to agentic systems, but the capability remains far less developed for Arabic, with its dialectal variation and morphological richness, than for English. The AISA-ArabicFC shared task (Nacar et al., 2026a) targets exactly this gap. Given an Arabic query in one of five dialects and a candidate set of tools, a system must (i) decide whether any tool applies (a binary call/no-call decision), (ii) select the correct function when one applies, and (iii) extract its arguments as structured key–value pairs. Track A scores function selection and argument extraction; Track B additionally requires an Arabic reasoning trace; Track C is a diagnostic ranking by the equal-weight macro-average of the organizers' per-dialect Overall score across the five dialects.

Two features of the task shaped our approach. First, the headline metric is Argument Exact-Match (ArgEM), a strict comparison of every predicted argument key and value; the organizers' baselines make clear that function selection is largely tractable while argument extraction is the open problem.[1] Second, the final ranking is determined not on the public development split but on a *blind* test set released only at the end of the competition. This is decisive: any gain that does not *transfer*—in particular, tuning that overfits the public split—is worthless. Our guiding question for every change was therefore not "does this raise the dev score?" but "will this transfer to the blind set?"

2 Task and Data

The AISA-ArabicFC dataset comprises Arabic queries across five dialect groups—Modern Standard Arabic (MSA), Levantine, Egyptian, Gulf, and Maghrebi—spanning eight service domains and a fixed inventory of structured tools (Nacar et al., 2026a). It follows the AISA architecture for agentic Arabic systems and its data-centric approach to reliable structured tool calling (Nacar et al., 2026c,b). Each instance presents the model with a user message and a candidate set of tools; the gold output is either a single tool call with its arguments, or an explicit no-call ("none") for negative instances. The public release provides a training split and a development split, with the blind test set released at the close of the competition. Training used 10,500 task positives plus 20,331 supplementary rows: 17,604 hash-deduplicated AISA-AR-FunctionCall positives and 2,727 dialect-balanced negatives from task/AISA-AR pools. Among the

[1] Under the released final scorer and gold, the organizers report AISA-Think at FnAcc 0.9813/ArgEM 0.6790 (Track A 0.8000) and prompted GPT-4o zero-shot at 0.9324/0.7442 (Track A 0.8195).

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combined positives, 5,600 copy-echo rows contained a gold argument verbatim in the user text; 2,800 were duplicated for $1.5\times$ effective weight. Build records preserve the criterion/count but not which 2,800 were duplicated. No development or test labels entered training.

Scoring uses a weighted composite per track. Writing FnAcc for function-name accuracy (exact match of the predicted function name, or "none" for negatives, which also penalizes hallucinated calls), ArgEM for argument exact-match over positive instances, and $\mathrm{ThinkRate}$ for the presence of an Arabic reasoning trace, the two track scores are:

$$\mathrm{A} = 0.40\,\mathrm{FnAcc} + 0.60\,\mathrm{ArgEM}, \quad (1)$$

$$\mathrm{B} = 0.30\,\mathrm{FnAcc} + 0.50\,\mathrm{ArgEM} + 0.20\,\mathrm{ThinkRate}. \quad (2)$$

On the development set ArgEM carried the largest weight and was furthest from saturation, so our engineering effort was directed there once function selection had saturated.

Data and scorer evolution. The task resources evolved through data v1.4 during the campaign. The organizers first released canonicalized data (v1.1) with a normalizing evaluator. Version 1.2 recovered unspecified gold arguments from execution traces in roughly 3% of training instances. Version 1.3 hardened the evaluator toward substance over surface form, normalizing product names, specialties and category aliases; v1.4 applied the organizers' annotation policy. The v1.2 and v1.3 changes each raised our measured scores without any change to our system, indicating that part of the measured difficulty was evaluation strictness rather than model error. Our locally re-implemented scorer was validated across v1.1–v1.3 and matched the official leaderboard to the fourth decimal on all eight submissions we made across the three evaluator versions.

2.1 System

Our system fine-tunes Qwen2.5-7B-Instruct (Team, 2025) using Low-Rank Adaptation (Hu et al., 2022) on the attention and MLP projection matrices.[2] We retained the natural dialect distribution; rebalancing it is examined and rejected in §4.5.

[2]Rank 64, $\alpha = 128$, dropout 0.05; learning rate 2×10^{-4}, effective batch 16, max sequence length 2048, 3-epoch schedule, seed 42.

Completion-only loss masking. The single most consequential design choice was to mask the prompt tokens from the training loss, computing the loss only over the completion (the reasoning trace and the structured call). We implemented this with a token-level response template so that gradient signal concentrates entirely on the function-calling output rather than being diluted across the instruction text. An earlier configuration that trained without this masking failed to reliably learn argument mappings it had nonetheless seen thousands of times in training—most visibly the mapping from Arabic currency and language names to their canonical ISO codes. Completion-only loss coincided with learning these mappings (§4).

Early models hallucinated calls on negative instances. We addressed this in training (reasoned Arabic abstention rationales) and at inference (the gold-free safety net, §3).

3 Methodology: Measurement-First Engineering

The distinctive part of our approach is not a single architectural trick but a discipline for deciding what to ship. We describe its three components.

3.1 A validated local scorer

The official evaluation is run only through the leaderboard, which returns a single score per submission. To make local iteration possible—and, more importantly, to make our decisions *checkable*—we re-implemented the scoring pipeline locally from the published evaluation code, mirroring its normalization rules and updating them at each evaluator revision. We validated this re-implementation against the official leaderboard on all eight of our submissions and obtained agreement to the fourth decimal place on every one (§2). From that point we treated the local scorer as a certified instrument: we could compute a submission's official score before uploading it.

3.2 Falsifiable predictions

We adopted a rule that no submission would be uploaded without a written, falsifiable prediction of its official score. This had two effects. It forced us to articulate *why* we expected a change to help, and by how much; and it gave us an immediate signal when an instrument—rather than a model—was misbehaving. A corollary rule, which we state plainly because it saved us twice, is that *an impossible number is evidence of a broken instrument,*

not of a broken model. On one occasion a sweep reported a function-name accuracy of 0.083 for a model whose predecessor had scored above 0.99; rather than concluding the model had collapsed, we treated the number as impossible and traced it to a parser that did not recognize the model's output format. The model was intact; the measurement was not.

3.3 The gold-free safety net

Post-processing on a public development set is dangerous precisely because the gold labels are visible: a rule can "fix" a development error by quietly memorizing which instances are negative, and such a rule will not transfer to the blind test, where no labels exist. We therefore constrained ourselves to rules that use only inference-time information. Our primary *per-sample candidate veto* forces out-of-list tools to "none." It was active on the blind test, firing on 64/1,125 rows (5.69%); without counterfactual gold we assign no separate score gain. Secondary N2 used only the abstention signal from dev-only V24, a Quran/customs/umrah hybrid (V14 tool decisions; V20F arguments on 14 rows). N2 was **not used on the blind test**: V24 had no test predictions.

A training-only enumeration rule. For `termination_type`, a training-only 13-entry Arabic-to-English map kept forms with $\geq$ 95% label consistency and ≥ 3 occurrences; ambiguous forms were untouched. It produced zero correct-to-wrong development flips, changed five predictions, corrected four, and raised Overall-B from 0.910 to 0.914.

4 Experiments and Results

4.1 Development-set progression

Table 1 traces our progression. The safety net lifted FnAcc from 0.95 to 0.996 with unchanged ArgEM; completion-only masking then lifted ArgEM from 0.732 to 0.778 and FnAcc to 1.0 (Overall-B 0.889). Gains came from learning canonical argument conventions the unmasked model had missed.

4.2 Error analysis

A field-level analysis of residual argument errors identified a categorical field whose gold had been reduced under v1.1 to canonical enumerated values while its *training* values remained free-form, so the model was learning an obsolete mapping. Remaining errors span recoverable mappings, copy-from-source extraction, and irreducible cases (relative dates, strict identifiers).

Configuration	FnAcc	ArgEM	B
Under the v1.1 scorer			
Base fine-tune	0.9505	0.732	0.851
+ safety net	0.9963	0.732	0.865
+ completion masking	**1.000**	**0.778**	**0.889**
Same model, re-scored under v1.3			
Pre-enumeration model	1.000	0.820	0.910
+ enumeration rule	**1.000**	**0.828**	**0.914**

Table 1: Development progression for the lost champion (§5.1). Upper block: v1.1 ablation. Lower: the same frozen predictions under v1.3, before/after the enumeration rule (§3.3). Blocks are not comparable because v1.3 normalizes previously rejected surface forms. ThinkRate is 1.0; the blind submission used a reconstruction (§4.4).

4.3 A targeted retraining negative result

Motivated by the error analysis, we corrected the obsolete categorical field in the training data (a `termination_type` training-label remap, 663 rows to five canonical values) together with an erroneous year in date fields (731 replacements), holding the recipe otherwise fixed. Applying both fixes as post-processing to the champion's frozen predictions gave a ceiling of ArgEM 0.826; the retrained model instead reached 0.740, against the champion's 0.778 (Overall-B 0.867 against 0.889; all v1.1). The targeted field improved—its field-level failures fell from 23 to 19, with no regression there—but nineteen exact-match rows were lost overall, spread across unrelated fields such as `num_persons` ($1 \rightarrow 23$ failures) and `days` ($10 \rightarrow 23$). We ran this once, with a single seed, so we cannot separate the effect from run-to-run variance; we report it because the direction was opposite to our pre-specified prediction. We did not promote this model.

4.4 Blind-test results

The submitted system is not the model of Table 1, whose weights were lost before test (§5.1). Checkpoint 5000 (epoch 2.38) was selected on a frozen hand-stratified 20-row development fingerprint: eight negatives (four N1 blind spots), all five enum-sensitive `termination_type` rows, and seven ordinary positives; the full gate covered four dialects including Levantine.[3] It matched

[3] Gate IDs: 0–4, 10, 14, 20, 50, 55, 84, 100, 125, 140, 164, 177, 200, 223, 417, 500.

the lost champion on $17/20$ under the June parser. The submitted strict-superset parser preserved all old successes and recovered one Qwen-native `<tool_call>` row with no regression; under v1.3 gold, checkpoint/champion were jointly correct on $18/20$ versus $19/20$. Hence the blind score is not same-model dev–test transfer, and the ArgEM drop ($0.828 \rightarrow 0.7994$) cannot be cleanly split between reconstruction mismatch and distribution shift. On the organizers' final blind-test leaderboard it obtains FnAcc 0.9831, ArgEM 0.7994, ThinkRate 1.0: Overall-B 0.8946 and Overall-A 0.8729, ranking 12th on both tracks. On Track C it scores 0.8547 (rank 15; leader 0.9075): Gulf 0.913, MSA 0.884, Levantine 0.864, Egyptian 0.836, Maghrebi 0.776; the official gap, computed from unrounded dialect scores, is 0.136.

4.5 Dialect-balancing hypothesis (rejected)

Our largest development dialect deficit was Levantine (ArgEM 0.746 versus 0.824 for MSA, pre-enumeration under v1.3), yet Levantine was already over-represented in training (16.9% versus 13.4% of positive development instances). Of sixteen attributable failures, twelve followed the same enumeration/date patterns as MSA, three reflected inconsistent gold currency values, and one was a dialect-comprehension error. Downsampling therefore addressed no supported cause; targeted augmentation was not tested. This was a diagnosis, not an experiment. Blind Track C reverses the Levantine–Egyptian ordering (0.864 versus 0.836), but for a reconstructed model under a different metric, so it does not validate the development diagnosis. Maghrebi is lowest (0.776) on 11 instances and should be read with that denominator.

5 Discussion and Lessons

Frozen-output estimates are not forecasts. We estimated our experiment's gain by applying the fix to a *frozen* model's outputs. Such an estimate bounds what the mechanism repairs in existing predictions and says nothing about what retraining does elsewhere.

In our experiment, retraining was diffuse. The corrections improved the fields they addressed, yet the model regressed overall through small losses across unrelated fields. We cannot generalize from one run, but holding the training recipe otherwise fixed let us *detect* the regression rather than ship it.

5.1 Checksum-authenticated identity confusion

A final lesson concerns the integrity discipline itself. Re-running the archived champion on 20 development instances and diffing against its frozen predictions failed at 11/20: the archived file was a rejected checkpoint from a parallel run, stored under the champion's name since an archival step six weeks earlier. Its SHA-256 had verified throughout—every check was correct, and every one attested a file that was never the right one.

We call this *checksum-authenticated identity confusion*: a checksum establishes that bits are unchanged, not that the intended artifact was hashed. Detecting it requires either an independent record binding the training run to the stored file, or evidence that the artifact *behaves* as the intended one. We had the latter unintentionally: development predictions retained for re-scoring served as an *incidental behavioral fingerprint*, and became the criterion for selecting among the retrained checkpoints. Behavioral fingerprints are established for distinguishing between known models (Yax et al., 2025; Wu et al., 2025); our observation is narrower and operational: a team's archived predictions already constitute one, and consulting them before an irreversible action is a check that hash verification cannot substitute for. Here they selected the system we submitted (§4.4).

6 Conclusion

Best-development B is 0.914; reconstruction ranks 12th on B (0.8946) and 15th on C (0.8547). Negative results and identity failure reinforce: measure first; verify before shipping.

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Limitations

The blind submission is not a clean same-model transfer measurement: the checkpoint selected for submission is not the model characterized in Table 1, and its agreement with the lost champion (18/20 versus 19/20 under the final v1.3 gold) means the drop between the development and blind ArgEM scores ($0.828 \rightarrow 0.7994$) cannot be cleanly attributed to reconstruction mismatch versus distribution shift (§4.4). The targeted-retraining negative result and the dialect-balancing diagnosis are each

based on a single run with a single seed, so neither the direction of the retraining regression nor the dialect diagnosis should be read as established beyond this run (§4.3, §4.5). We did not train a same-base control without LoRA, so the contribution of LoRA fine-tuning relative to the base model is not isolated. The two blocks of Table 1 are scored under different evaluator versions and are not directly comparable; part of the apparent improvement between them reflects evaluator changes rather than model changes (§2). Finally, the 2,800 duplicated rows in the training mixture were selected by an automated copy-echo criterion whose build records preserve the criterion and count but not the specific rows selected, so that step of data preparation is not exactly reproducible from the released description alone (§2).

Ethics Statement

This work uses the AISA-ArabicFC shared-task dataset as released by the organizers; we did not collect additional data or conduct human-subjects research. Our system is a research artifact evaluating Arabic function-calling reliability and carries no dual-use risk beyond that of any general-purpose tool-calling assistant. All experiments were run on the publicly released training and development splits, and the blind test set was scored only through the organizers' official leaderboard.

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